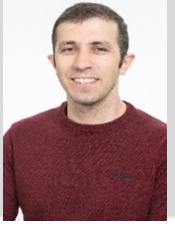
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LinkedIn , Google Scholar, GitHub



Research Highlights

- **AI-guided biosensor design**: using machine learning, deep learning and graph neural networks to design DNA sequences for optical nanotube sensors.
- **Nanobiosensors**: engineering nanomaterial-based sensors, such as DNA-wrapped carbon nanotubes, to detect molecules like glucose, cancer biomarkers, nitric oxide and mycotoxins.
- **High-throughput and autonomous experimentation:** building automated screening and data pipelines, and moving toward self-driving labs that close the design, test and learn loop.
- **Functional materials and rheology:** designing magnetorheological fluids and superhydrophobic materials, and modeling their behavior with experimental design and machine learning.

Education and Training

Invited Guest Scientist and Postdoctoral Researcher, EPFL (École Polytechnique Fédérale de Lausanne), School of Architecture, Civil and Environmental Engineering (ENAC), Ludwig group, Lausanne, Switzerland (2025.03-now)

- Designing AI-enhanced nanobiosensor arrays for disease diagnostics and health monitoring

Postdoctoral Researcher in Chemistry and Chemical Engineering, EPFL (École Polytechnique Fédérale de Lausanne), Laboratory of Nanobiotechnology (LNB) and Ludwig group, Lausanne, Switzerland (2022.12-2025.03)

- Prepared and characterized ssDNA-SWCNT complexes and investigated sequence-dependent optical responses using fluorescence spectroscopy, UV-Vis-NIR spectroscopy, and NIR microscopy.
- Developed closed-loop AI discovery platform combining unsupervised clustering, deep learning prediction, and automated high-throughput screening: reducing sensor discovery time from ~2 years to under 6 months.
- Developed automated 96/384-well screening workflows integrating liquid handling, optical measurements, and reproducible data processing to compare candidate sensor formulations.
- Coordinated computational and experimental work with EPFL researchers and CSEM collaborators, using screening results to guide subsequent candidate selection and validation.
- Mentored Bachelor's-Ph.D. researchers in **experimental design, data analysis, and quantitative interpretation**.

Ph.D. Exchange in Chemistry and Chemical Engineering, EPFL, Laboratory of Nanobiotechnology (LNB), Lausanne, Switzerland (2021-2022)

- Developed an ML model to predict ssDNA-SWCNT nanobio-optical sensor responses for mycotoxin detection.

Ph.D. in Chemical Engineering, University of Tehran (UT), Tehran, Iran (2017-2022)

- Awarded a full Ph.D. scholarship by the University of Tehran.
- Excelled in candidacy exams in mass transfer, heat transfer, thermodynamics, and reactor design; graduated with a GPA of 3.83/4 and an A-grade thesis.
- Applied machine learning techniques to enhance microalgae growth, nutrient removal, and lipid extraction for bioenergy.
- Improved wastewater treatment performance using superhydrophobic micro- and nanomagnetic particles.
- Supervised and mentored students from Bachelor's to Master's level, guiding research design, and experiments

M.Sc. in Chemical Engineering, Iran University of Science and Technology (IUST), Tehran, Iran (2013 –2015)

- Ranked 2nd at the College of Engineering; graduated with a GPA of 4.0/4.0 and an A-grade thesis.
- Specialized in modeling, simulation, and process optimization for chemical engineering applications.
- Served as a Teaching Assistant for courses including Process Optimization, Computational Fluid Dynamics (CFD), Applied Mathematics, Fluid Mechanics, and Differential Equations.

Bachelor of Science in Chemical Engineering, University of Tehran (UT), Tehran, Iran (2009 –2013)

- Ranked 3rd in Chemical Engineering; graduated with a GPA of 3.81/4.0.

Patents

- **Y. Rabbani (60%), A. A. Boghossian, J. Bregy, optimizing optical nano-biosensors made of single-walled carbon nanotubes wrapped in single-stranded DNA sequences for detection of molecules such as glucose and cancer biomarkers (AI-enabled discovery platform), (EP4742248A1).**
- **Y. Rabbani** (80%), A. Hamta, Sustainable magnetorheological fluid synthesis with proper rheological properties for industrial applications using nanoparticles and stabilizing additives. Iranian Patent No. 98516, 2017.

Books

- S. Behjati, **Y. Rabbani,** A. A. Boghossian, Bioengineering DNA for Carbon Nanotube Sensors, *Handbook of Carbon Nanomaterials*, Volumes 2, Chapter 8, 421-454, 2025.
- **Y. Rabbani**, S.M Sadeghi, Process Optimization in Chemical Engineering, University of Science and Technology, 2020.

Publications by Research Theme (Google Scholar)

1. Nano Biosensor Engineering

1. **Y. Rabbani,** J. Bregy, … , A.A. Boghossian**, "Discovery of a DNA-based Optical Nanotube Sensor for Glucose Using Clustering and Deep Learning Algorithms", *bioRxiv,* 2025.**
2. **Y. Rabbani** and A.A. Boghossian, "Advancements in the Surface Functionalization of Carbon Nanotubes for Optoelectronic Modulation and Sensing Applications", The Electrochemical Society Interface 3471,2025. **Featured Articles**
3. V. Zubkovs, L. Belcastro, S. H. Sajjadi, **Y. Rabbani,** A. Rastiniemi, …, S. Cattaneo, A. A. Boghossian, V. Basoli**, "Development of an Electronics-Free Optical Sensor Using Single-Walled Carbon Nanotubes for Monitoring Nitric Oxide in Inflammatory Responses", *Biosensors and Bioelectronics*, 293, 118092, 2026**.
4. **Y. Rabbani**, S. Behjati, B. Lambert, S. H. Sajjadi, M. Shariaty-Niassar et al., "Prediction of Mycotoxin Response of DNA-Wrapped Nanotube Sensor with Machine Learning", *bioRxiv*, 2023.09.07.556334, 2023.
5. **Y. Rabbani,** and S. Behjati (same contribution), B. Rousseau, S. H. Sajjadi, N. Sharif, A. A. Boghossian, A new clustering approach for the engineering of the ssDNA-SWCNT sensor for OPG cancer biomarker detection, 2025.
6. **Y. Rabbani**, S. H. Sajjadi, S. Behjati, N. Sharif, A. A. Boghossian, Machine learning Engineering of the ssDNA-SWCNTs Sensor for Nitric Oxide Detection, 2025.
7. B. Rousseau, **Y. Rabbani**, S. Behjati, S. H. Sajjadi, A. A. Boghossian, Machine learning Engineering of the ssDNA hybridization on the surface of SWCNTs, 2025.

2. AI/ML-Driven Process and Bioprocess Optimization

1. **Y. Rabbani,** M. Shirvani, S. H. Hashemabadi, M. Keshavarz**, "Application of Artificial Neural Networks and Support Vector Regression Modeling in Prediction of Magnetorheological Fluid Rheometry,"** *Colloids and Surfaces A: Physicochemical and Engineering Aspects*, 520, 268-278, 2017.
2. N. Hajinajaf, A. Fallahi, **Y. Rabbani**, O. Tavakoli, M. H. Sarrafzadeh, "Integrated CO_2 Capture and Nutrient Removal by Microalgae Chlorella vulgaris and Optimization Using Neural Network and Support Vector Regression," *Waste and Biomass Valorization*,13, 12,4749-4770, 2022.
3. N. Hajinajaf, **Y. Rabbani**, A. Mehrabadi, O. Tavakoli, "Experimental and Modeling Assessment of Large-Scale Cultivation of Microalgae Nannochloropsis sp. PTCC 6016 to Reach High Efficiency Lipid Extraction," *International Journal of Environmental Science and Technology*, 19, 6,5511-5528, 2022.
4. **Y. Rabbani**, M. Shariaty-Niassar, S. A. S. Ebrahimi, "The Optimization Effect of Different Parameters on the Super Hydrophobicity of Prickly-Shaped Carbonyl Iron Particles," *RSC Advances*, 12, 20, 12760-12772, 2022.
5. J. Maroofi, S. H. Hashemabadi, **Y. Rabbani**, "Investigation of the Chain Formation Effect on Thermal Conductivity of Magnetorheological Fluids," *Journal of Thermophysics and Heat Transfer*, 34, 1,3-12, 2020.
   Y. Rabbani, N. Hajinajaf, O. Tavakoli, "An Experimental Study on Stability and Rheological Properties of Magnetorheological Fluid Using Iron Nanoparticle Core–Shell Structured by Cellulose," *Journal of Thermal Analysis and Calorimetry*, 135,1687-1697, 2019.

3. Process Modeling, Transport Phenomena, and Materials Engineering

1. **Y. Rabbani**, M. Shariaty-Niassar, S. A. S. Ebrahimi, "The Effect of Superhydrophobicity of Prickly Shape Carbonyl Iron Particles on the Oil-Water Adsorption," *Ceramics International*, 47, 20, 28400-28410, 2021.
2. S. H. Sajjadi, S. J. Wu, **Y. Rabbani**, V. Zubkovs, H. Ahmadzadeh et al., "Micropreparative Gel Electrophoresis for Purification of Nanoscale Bioconjugates," *Bioconjugate Chemistry*, 35, 2,154-163, 2024.
3. S. Behjati, L. H., B. Rousseau, **Y. Rabbani**, H. Wang, S. Dehury, R. Di Costanzo, S. H. Sajjadi, D. Constam, A. A. Boghossian. "Enabling pH resilience of an optical glutathione sensor by distinguishing enthalpy- and entropy-driven interactions", *Small Methods*, 2025. (Under Review)
4. **Y. Rabbani**, N. Hajinajaf, M.S. Niassar, "The Effect of Microparticles/Nanoparticles Surface Modification on the Magnetorheological Fluid Properties:A Review," *Intelligent Material Systems and Structures*, 34,15,1715, 2023.
5. H. Shayesteh, M. S. Khosrowshahi, H. Mashhadimoslem, F. Maleki, **Y. Rabbani**, et al., "Durable Superhydrophobic/Superoleophilic Melamine Foam Based on Biomass-Derived Porous Carbon and Multi-Walled Carbon Nanotube for Oil/Water Separation," *Scientific Reports*,13,1, 4515, 2023.
6. **Y. Rabbani**, H. Shayesteh, N. Haghshenas, M. Safarzadeh Khosrowshahi, "Investigation of Grafting Silane Coupling Agents on Superhydrophobicity of Carbonyl Iron/SiO_2 Particles for Efficient Oil/Water Mixture and Emulsion Separation," *Scientific Reports*, 13, 1,788, 2023.
7. **Y. Rabbani**, M. Ashtiani, S. H. Hashemabadi, "An Experimental Study on the Effects of Temperature and Magnetic Field Strength on the Magnetorheological Fluid Stability and MR Effect," *Soft Matter*, 11,22,4453-4460, 2015

Peer-reviewed conference articles

1. **Y. Rabbani**, S. H. Hashemabadi, S. Golshan, "CFD-DEM Three-dimensional Simulation of Magnetorheologiccal Fluids", *The 9th International Chemical Engineering Congress and Exhibition (IChEC 2015)*, 2015,
2. **Y. Rabbani**, O. Tavakoli ,"Investigation of Heat and Mass Transfer in Magnetorheological Fluids", *Heat and Mass Transfer in Iran*, 2017, (In Persian).
3. M. Homayoonfal, **Y. Rabbani**, M. Ezati, M. S.hariati-Niassar ,"Industrial synthesis of ZnO nanoparticles: study on the synthesis method, characteristics, and suggestive proposal for application in the field of polymeric materials", *The 8th International Chemical Engineering Congress and Exhibition (IChEC)*, Iran, 2014.
4. M. Homayoonfal, **Y. Rabbani**, M. Ezati, M. S.hariati-Niassar ,"Industrial synthesis of ZnO nanoparticles: study on the synthesis method, characteristics, and suggestive proposal for application in biological filtration medium", *IWA Balkan Young Water Professionals*, 2015.

Conference Proceedings and Abstracts

1. **Y. Rabbani**, J. Bregy, B. Rousseau, S. H. Sajjadi, L. D. Benedittis, S. Behjati, M. Mouhib, S. Dehury, A. A. Boghossian, "Designing Optical Nanobiosensors for Glucose Detection Based on Clustering and Deep Learning", *Biomedical Photonics Network Meeting,* Zürich Switzerland, 2024,
2. **Y. Rabbani**, S. Behjati, B. Lambert, …, A. A. Boghossian, "Prediction of mycotoxin response of DNA-wrapped nanotube sensor with machine learning", *Swiss Symposium in Point-of-Care Diagnostics,* Zürich Switzerland, 2024.
3. **Y. Rabbani**, S. Behjati, B. Lambert, S. Sajjadi, M.S. -Niassar, A. A. Boghossian ,"Prediction of mycotoxin response of DNA-wrapped nanotube sensor with machine learning", *ChemOnTubes Conference in Strasbourg France*, 2024.
4. S. Behjati, L. Huang, B. Rousseau, H. Wang, **Y. Rabbani**, R. Di Costanzo, S. H. Sajjadi, A. A. Boghossian ,"Engineering pH Resilience in Optical Nanotube Sensors for Biomedical Applications", *ECS (The Electrochemical Society), 245th ECS Meeting in San Francisco USA*, 2024.
5. S. H. Sajjadi, **Y. Rabbani**, S. Behjati, V. Zubkovs, S. Cattaneo, V. Basoli, S. Grad, A. A. Boghossian , "Directed Evolution and Machine Learning for Improved ssDNA-SWCNTs Sensor for Nitric Oxide Detection", *244th ECS Meeting in Gothenburg Sweden,* 2023. **(Invited)**
6. **Y. Rabbani**, S. Behjati, B. Lambert, S. H. Sajjadi, M. Shariaty-Niassar and A. A. Boghossian , "Prediction of Mycotoxin Response of DNA-Wrapped Nanotube Sensor with Machine Learning", *244th ECS Meeting in Gothenburg,* 2023.
7. **Y. Rabbani**, S. Behjati, B. Lambert, S. H. Sajjadi, M. S. Niassar and A. A. Boghossian , "Prediction of Mycotoxin Response of DNA-Wrapped Nanotube Sensor with Machine Learning", *243th ECS Meeting in Boston USA*, 2023.
8. **Y. Rabbani**, S. Behjati, B. Lambert ,…, A. A. Boghossian,"Application of machine learning on DNA-SWCNT nano biosensor to detect of different analytes", *Biomedical Photonics Network Meeting,* Lausanne Switzerland, 2022.

9. **Y. Rabbani**, S. Behjati, B. Lambert, S. Sajjadi, M. S. Niassar, A. A. Boghossian ,"Prediction of mycotoxin response of DNA-wrapped nanotube sensor with machine learning", *Swiss chemical society (SCS),* Zurich Switzerland, 2022.
10. **Y. Rabbani**, H. Hashemabadi, S. Golshan, A. Daneshgar , "CFD-DEM Simulation of Magnetorheological Fluids", *15th International Conference on Electrorheological Fluids and Magnetorheological Suspensions*, Incheon, Korea, 2016.

Teaching Experience

2012 – Present

Teaching Assistant and Instructor, University of Tehran (UT), Iran University of Science and Technology (IUST), and École Polytechnique Fédérale de Lausanne (EPFL)

Undergraduate Courses: Fluid Mechanics, Differential Equations, Applied Mathematics in Chemical Engineering.

Graduate Courses: Computational Fluid Dynamics (CFD), Process Optimization in Chemical Engineering, Machine Learning.

- Delivered lectures, led discussion sessions, graded assignments, and provided one-on-one academic support, including guidance on course projects, research activities, and report writing

Mentorship and Supervision

- **Undergraduate Students:** Supervised 10+ Bachelor's students.
- **Master's Students:** Supervised 5+ Master's students.
- **PhD Students:** Co-supervised and mentored one doctoral researcher at EPFL.
- **Research Project:** Projects spanned chemical engineering, nanomaterials, machine learning, and biosensors, with training in nanobiosensor engineering, AI-assisted biosensor discovery, experimental design, data analysis, process optimization, and wastewater treatment.
- **Institutions:** Mentored students across the University of Tehran, Iran University of Science and Technology, and EPFL, with an emphasis on inclusive mentoring for international and first-generation students.

Leadership and Professional Service

- **Editorial Roles:** Topic Editor, Frontiers in Environmental Chemistry; Guest Editor, Journal of Applied Sciences, providing leadership in evaluating and curating peer-reviewed research.
- **Journal Reviewer:** Chemical Engineering Journal, Ceramics International, Journal of Magnetism and Magnetic Materials, Environmental Science, International Journal of Heat and Fluid Flow, Journal of Heat and Mass Transfer, RSC
- **Conference Leadership:** Chair of Synthesis and Characterization of Carbons section at the 244th ECS Meeting
- **Team Leadership:** Led a team to translate scientific concepts into market-ready solutions.
- **Competition Mentorship:** Led the Chem-E-Car team to a national competition.
- **Student Association Supervision:** Supervised the Association of Chemical Engineering at the University of Tehran.

Technical Skills and Lab Instruments

Laboratory Instruments and Techniques

- Absorbance and fluorescence spectroscopy (UV-Vis-NIR, fluorescence), NIR optical microscopy/imaging
- Nanomaterials synthesis and optical nanosensor engineering (ssDNA-SWCNT)
- Materials and analytical characterization: FT-IR, XRD, TGA, DLS, VSM, SEM, and TEM

Automation and data pipelines:

- LabVIEW, instrument control, automated data processing, and high-throughput experimentation.

Scientific ML:

- PyTorch, TensorFlow, scikit-learn; GNNs, CNNs, random forests, SVMs, active-learning model-guided optimization.

Molecular and sequence informatics:

- RDKit, SMILES molecular representation, molecular descriptors, k-mer analysis, MAFFT alignment, RNA secondary-structure prediction (ViennaRNA), and clustering.

Scientific computing:

- Python, NumPy, pandas, Matplotlib, MATLAB, Git, and Conda.

Mechanistic modeling:

- physics-informed and hybrid modeling, CFD, DEM, Aspen HYSYS, and OpenFOAM.